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(Pages : 3)

Name.....

Reg. No.....

SIXTH SEMESTER (CBCSS—UG) DEGREE EXAMINATION, MARCH 2025

Mathematics

MTS 6B 10—REAL ANALYSIS

(2019 Admission onwards)

Time : Two Hours and a Half

Maximum : 80 Marks

Section A*Answer any number of questions.**Each question carries 2 marks.**Ceiling is 25.*

1. Prove that polynomial functions are continuous on $\mathbb{R}$.
2. Give an example of uniformly continuous function which is not a Lipschitz function.
3. State Weierstrass approximation theorem.
4. Prove that every constant function on $[a, b]$ is Riemann integrable on $[a, b]$.
5. Suppose f and g are in $\mathcal{R}[a, b]$. If $f(x) \leq g(x)$ for all $x \in [a, b]$, prove that $\int_a^b f \leq \int_a^b g$.
6. State substitution theorem and use this to evaluate $\int_1^4 \frac{\sin \sqrt{t}}{\sqrt{t}} dt$.
7. Prove that rational numbers in $[0, 1]$ is a null set.
8. If $f \in \mathcal{R}[a, b]$, Prove that $|f| \in \mathcal{R}[a, b]$ and $\left| \int_a^b f \right| \leq \int_a^b |f|$.
9. Define uniform norm on a set of bounded functions.
10. Evaluate $\lim (e^{-nx})$ for $x \in \mathbb{R}, x \geq 0$.

Turn over

11. State and prove Weierstrass M test.
12. Evaluate $\int_1^{\infty} \frac{dx}{x^2 + 1}$.
13. Find P.V. $\int_{-\infty}^{\infty} x dx$.
14. Prove that $\int_2^{\infty} \frac{1}{\ln(x)} dx = \infty$.
15. Prove that $\Gamma(p)$ is strictly convex for $0 < p < \infty$.

Section B

Answer any number of questions.

Each question carries 5 marks.

Ceiling is 35.

16. Prove that Dirichlet's function is discontinuous on $\mathbb{R}$.
17. State and prove preservation of intervals theorem.
18. If $f : [a, b] \rightarrow \mathbb{R}$ is continuous on $[a, b]$, then prove that $f \in \mathcal{R}[a, b]$.
19. State and prove Taylor's theorem with the remainder.
20. Let (f_n) be sequence of continuous functions on a set $A \subseteq \mathbb{R}$ and suppose that (f_n) converges uniformly on A to a function $f : A \rightarrow \mathbb{R}$. Prove that f is continuous on A.
21. Prove that for all $q \leq -1$, $\int_0^1 x^q e^{-x} dx$ diverges.
22. Prove that $\int_0^{\infty} \sqrt{y} e^{-y^2} dy = \frac{1}{2} \Gamma(3/4)$.
23. Prove that $B(p, q) = \frac{\Gamma(p) \Gamma(q)}{\Gamma(p+q)}$ for all $p > 0$ and $q > 0$.

Section C

*Answer any **two** questions.
Each question carries 10 marks.
Maximum 20 marks.*

24. State and prove location of roots theorem.
25. State and prove second form of Fundamental theorem of calculus.
26. State and prove Cauchy criterion for uniform convergence.
27. For any $p > 0$, fixed, Prove that $\lim_{0 < q \rightarrow \infty} q^p B(p, q) = \Gamma(p)$.

(2 × 10 = 20 marks)